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e-mail: contact@mathsmanthan.co.in | Website:- mathsmanthan.co.in

Subject: Mathematics **CBSE CLASS 10 – PERIODIC ASSESSMENT 11** **Time:** 60 Minutes
Class: Class X (CBSE) **Chapter 11: Areas Related to Circles** **MM:** 30

Document Type: Official Step-by-Step Marking Scheme **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper physical units.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

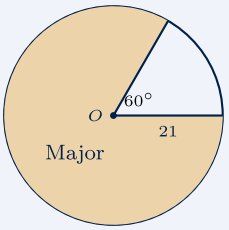
SECTION A: MULTIPLE CHOICE

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Q1. If a sector of a circle of radius 21 cm has central angle 60°, then the area of the corresponding major sector is: $\left(\text{Use } \pi = \frac{22}{7}\right)$ [1]

(A) 1155 cm^2 (B) 231 cm^2 (C) 1386 cm^2 (D) 924 cm^2

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]



Step 1: Total Area and Major Sector Angle [0.5 Mark]

Radius $r = 21$ cm. Central angle of major sector:

$$\theta' = 360^\circ - 60^\circ = 300^\circ$$

Step 2: Area Calculation [0.5 Mark]

$$\text{Area of major sector} = \frac{\theta'}{360^\circ} \times \pi r^2 = \frac{300^\circ}{360^\circ} \times \frac{22}{7} \times 21 \times 21$$

$$= \frac{5}{6} \times 22 \times 3 \times 21 = \frac{5}{6} \times 1386 = 5 \times 231 = 1155 \text{ cm}^2$$

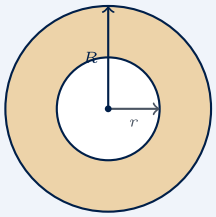
Correct Option: (A) 1155 cm^2

Area of Major Sector = **1155** cm^2

Q2. If the ratio of the areas of two concentric circles is 1 : 4 and their boundary ring has width 7 cm, then the radius of the inner circle is: [1]

(A) 7 cm (B) 14 cm (C) 3.5 cm (D) 21 cm

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]



Step 1: Ratio of Radii [0.5 Mark]

Let inner radius be r and outer radius be R.

$$\frac{\text{Area}_1}{\text{Area}_2} = \frac{\pi r^2}{\pi R^2} = \frac{1}{4} \implies \frac{r}{R} = \frac{1}{2} \implies R = 2r$$

Step 2: Width of Ring [0.5 Mark]

$$\text{Width } w = R - r = 2r - r = r = 7 \text{ cm}$$

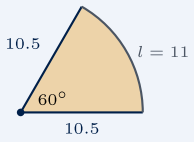
Correct Option: (A) 7 cm

Radius of inner circle $r = \mathbf{7}$ cm

Q3. The perimeter of a sector of a circle of radius 10.5 cm with central angle 60° is: $\left(\text{Use } \pi = \frac{22}{7}\right)$ [1]

(A) 32 cm (B) 11 cm (C) 21 cm (D) 42 cm

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]



Step 1: Arc Length Calculation [0.5 Mark]

$$l = \frac{\theta}{360^\circ} \times 2\pi r = \frac{60^\circ}{360^\circ} \times 2 \times \frac{22}{7} \times 10.5 = \frac{1}{6} \times 66 = 11 \text{ cm}$$

Step 2: Total Sector Perimeter [0.5 Mark]

$$\text{Perimeter} = l + 2r = 11 + 2(10.5) = 11 + 21 = 32 \text{ cm}$$

Correct Option: (A) 32 cm

Perimeter of sector = **32** cm

Q4. In a circle of radius r, the area of the triangle formed by two radii with central angle 120° is given by: [1]

(A) $\frac{\sqrt{3}}{4}r^2$ (B) $\frac{1}{2}r^2$ (C) $\frac{\sqrt{3}}{2}r^2$ (D) $\frac{1}{4}r^2$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Step 1: Triangle Area Formula [0.5 Mark]

For $\triangle OAB$ with sides $OA = OB = r$ and included angle $\angle AOB = \theta$:

$$\text{Area}(\triangle OAB) = \frac{1}{2} r^2 \sin\theta$$

Step 2: Value for $\theta = 120^\circ$ [0.5 Mark]

$$\sin 120^\circ = \sin(180^\circ - 60^\circ) = \sin 60^\circ = \frac{\sqrt{3}}{2}$$

$$\text{Area}(\triangle OAB) = \frac{1}{2} r^2 \left(\frac{\sqrt{3}}{2}\right) = \frac{\sqrt{3}}{4} r^2$$

Correct Option: (A) $\frac{\sqrt{3}}{4} r^2$

$$\boxed{\text{Area of Triangle} = \frac{\sqrt{3}}{4}\mathbf{r^2}}$$

Q5. If the perimeter of a circle is decreased by 50%, then its area is decreased by: [1]

(A) 25% (B) 50% (C) 75% (D) 100%

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Step 1: Change in Radius [0.5 Mark]

$C = 2\pi r$. If C decreases by 50%, then new perimeter $C' = \frac{1}{2}C \implies r' = \frac{1}{2}r$.

Step 2: Change in Area [0.5 Mark]

Original area $A = \pi r^2$. New area $A' = \pi(r')^2 = \pi\left(\frac{r}{2}\right)^2 = \frac{1}{4}\pi r^2 = 0.25A$.

$$\text{Decrease in area} = \frac{A - A'}{A} \times 100\% = \frac{A - 0.25A}{A} \times 100\% = 75\%$$

Correct Option: (C) 75%

$$\boxed{\text{Area is decreased by } \mathbf{75}\%}$$

Q6. **Assertion (A):** If the circumference of a circle is 176 cm, then its radius is 28 cm.
Reason (R): The circumference C of a circle of radius r is given by $C = 2\pi r$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(C) (A) is true but (R) is false.

(D) (A) is false but (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Step 1: Verification of Assertion (A) [0.5 Mark]

$$C = 2\pi r = 176 \implies 2 \times \frac{22}{7} \times r = 176 \implies \frac{44}{7} r = 176 \implies r = \frac{176 \times 7}{44} = 28 \text{ cm}$$

Hence, Assertion (A) is **True**.

Step 2: Verification of Reason (R) [0.5 Mark]

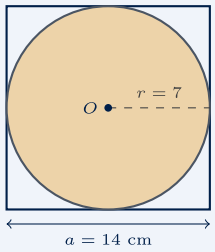
Reason (R) states the standard geometric formula $C = 2\pi r$, which directly calculates and validates Assertion (A). Hence, (R) is **True** and correctly explains (A).

Correct Option: **(A) Both (A) and (R) are true and (R) is the correct explanation of (A).**

$$\boxed{\textbf{Option (A) is correct}}$$

Q7. If the area of a circle inscribed in a square is 154 cm^2, find the area of the square. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [2]

▶ **STEP-BY-STEP SOLUTION:** [2 Marks]



Step 1: Finding Radius of Inscribed Circle [1 Mark]

Let the radius of the inscribed circle be r cm.

$$\text{Area of circle} = \pi r^2 = 154 \text{ cm}^2$$

$$\frac{22}{7} r^2 = 154 \implies r^2 = \frac{154 \times 7}{22} = 7 \times 7 = 49 \implies r = 7 \text{ cm}$$

Step 2: Side and Area of Square [1 Mark]

For a circle inscribed in a square of side a:

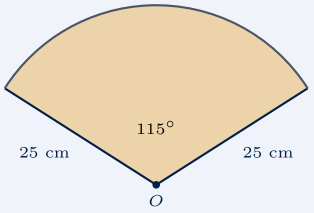
$$\text{Side of square } a = \text{Diameter } = 2r = 2 \times 7 = 14 \text{ cm}$$

$$\text{Area of square} = a^2 = 14^2 = 196 \text{ cm}^2$$

$$\boxed{\text{Area of the square} = \mathbf{196} \text{ cm}^\mathbf{2}}$$

Q8. A car wiper blade of length 25 cm sweeps through an angle of 115°. Calculate the area cleaned in one sweep of this blade. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [2]

▶ **STEP-BY-STEP SOLUTION:** [2 Marks]



Step 1: Formula and Given Parameters [1 Mark]

Length of blade (radius) $r = 25$ cm, Swept angle $\theta = 115°$.

$$\text{Area swept} = \frac{\theta}{360°} \times \pi r^2$$

Step 2: Evaluation [1 Mark]

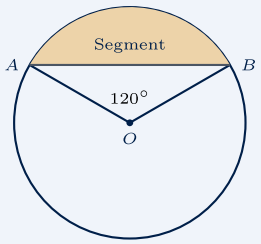
$$\text{Area} = \frac{115°}{360°} \times \frac{22}{7} \times 25 \times 25 = \frac{23}{72} \times \frac{22}{7} \times 625$$

$$= \frac{23 \times 11 \times 625}{36 \times 7} = \frac{158125}{252} \approx 627.48 \text{ cm}^2$$

$$\boxed{\text{Area cleaned in one sweep} = \frac{158125}{252} \text{ cm}^2 \approx \mathbf{627.48} \text{ cm}^\mathbf{2}}$$

Q9. A chord of a circle of radius 10.5 cm subtends an angle of 120° at the centre. Find the area of the corresponding minor segment. $\left(\text{Use } \pi = \frac{22}{7}, \sqrt{3} = 1.732\right)$ **[3]**

▶ **STEP-BY-STEP SOLUTION:** **[3 Marks]**



Step 1: Area of Sector OAB **[1 Mark]**

Radius $r = 10.5$ cm $= \frac{21}{2}$ cm, Central angle $\theta = 120°$.

$$\text{Area}_{\text{sector}} = \frac{120°}{360°} \times \frac{22}{7} \times \left(\frac{21}{2}\right)^2 = \frac{1}{3} \times \frac{22}{7} \times \frac{441}{4} = \frac{231}{2} = 115.5 \text{ cm}^2$$

Step 2: Area of $\triangle OAB$ **[1 Mark]**

$$\text{Area}_{\triangle OAB} = \frac{1}{2} r^2 \sin 120° = \frac{1}{2}\left(\frac{21}{2}\right)^2 \times \frac{\sqrt{3}}{2} = \frac{441\sqrt{3}}{16}$$

$$= \frac{441 \times 1.732}{16} = \frac{763.812}{16} \approx 47.74 \text{ cm}^2$$

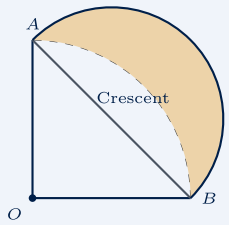
Step 3: Area of Minor Segment **[1 Mark]**

$$\text{Area}_{\text{segment}} = \text{Area}_{\text{sector}} - \text{Area}_{\triangle OAB} = 115.5 - 47.74 = 67.76 \text{ cm}^2$$

Area of Minor Segment = **67.76** cm^2

Q10. In a circle of radius 14 cm, a sector has central angle 90°. A semicircle is drawn on the chord of this sector as diameter. Find the area of the shaded crescent region. $\left(\text{Use } \pi = \frac{22}{7}\right)$ **[3]**

▶ **STEP-BY-STEP SOLUTION:** [3 Marks]



Step 1: Chord AB and Area of Semicircle [1 Mark]

In quadrant OAB, $OA = OB = 14$ cm, $\angle AOB = 90°$.

$$\text{Chord } AB = \sqrt{14^2 + 14^2} = 14\sqrt{2} \text{ cm} \implies \text{Radius } R = 7\sqrt{2} \text{ cm}$$

$$\text{Area of semicircle on } AB = \frac{1}{2}\pi R^2 = \frac{1}{2} \times \frac{22}{7} \times 98 = 154 \text{ cm}^2$$

Step 2: Area of Minor Segment on Chord AB [1 Mark]

$$\text{Area}_{\text{segment}} = \text{Area(quadrant)} - \text{Area}(\triangle OAB)$$

$$= \frac{1}{4} \times \frac{22}{7} \times 14^2 - \frac{1}{2} \times 14 \times 14 = 154 - 98 = 56 \text{ cm}^2$$

Step 3: Area of Shaded Crescent (Lune) [1 Mark]

$$\text{Area}_{\text{crescent}} = \text{Area}_{\text{semicircle}} - \text{Area}_{\text{segment}} = 154 - 56 = 98 \text{ cm}^2$$

(Note: By Hippocrates' Theorem, the crescent area equals Area$(\triangle OAB) = 98$ cm^2.)

Area of the shaded crescent = **98** cm^2

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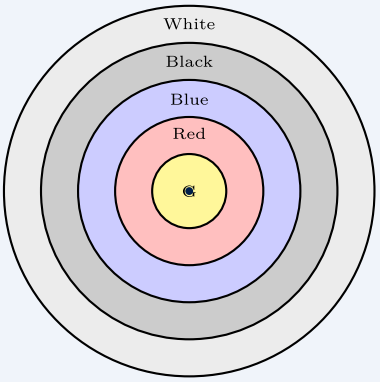
Q11. An Olympic archery target consists of five concentric scoring areas: Gold ($d = 21$ cm), Red, Blue, Black, and White, where each outer band is 10.5 cm wide. **[5]**

(a) Calculate the radius of each concentric boundary.

(b) Calculate the area of each of the five individual scoring regions. $\left(\text{Use } \pi = \frac{22}{7}\right)$

(c) Prove that the areas of the regions are in the arithmetic progression ratio $1 : 3 : 5 : 7 : 9$.

► STEP-BY-STEP SOLUTION: [5 Marks]



(a) Radii of Concentric Boundaries [1 Mark]

Given inner diameter of Gold $d = 21$ cm $\implies r_1 = 10.5$ cm. Band width $w = 10.5$ cm.

$r_1 = 10.5$ cm, $r_2 = 21$ cm, $r_3 = 31.5$ cm, $r_4 = 42$ cm, $r_5 = 52.5$ cm

(b) Areas of Individual Scoring Regions [2.5 Marks]

- Area(Gold) $= \pi r_1^2 = \frac{22}{7} \times 10.5^2 = 346.5$ cm^2
- Area(Red) $= \pi(21^2 - 10.5^2) = \frac{22}{7} \times 330.75 = 1039.5$ cm^2
- Area(Blue) $= \pi(31.5^2 - 21^2) = \frac{22}{7} \times 551.25 = 1732.5$ cm^2
- Area(Black) $= \pi(42^2 - 31.5^2) = \frac{22}{7} \times 771.75 = 2425.5$ cm^2
- Area(White) $= \pi(52.5^2 - 42^2) = \frac{22}{7} \times 992.25 = 3118.5$ cm^2

(c) Proof of AP Ratio $1 : 3 : 5 : 7 : 9$ [1.5 Marks]

Since $r_n = n \times 10.5 = nr_1$:

$$\text{Area}_n = \pi(r_n^2 - r_{n-1}^2) = \pi r_1^2[n^2 - (n-1)^2] = \text{Area}_1 \times (2n - 1)$$

For $n = 1, 2, 3, 4, 5$, $(2n - 1) = 1, 3, 5, 7, 9$.

$$\text{Ratio} = 1 : 3 : 5 : 7 : 9 \quad \textbf{(Hence Proved)}$$

Areas: Gold = 346.5, Red = 1039.5, Blue = 1732.5, Black = 2425.5, White = 3118.5 cm^2; Ratio = **1 : 3 : 5 : 7 : 9**

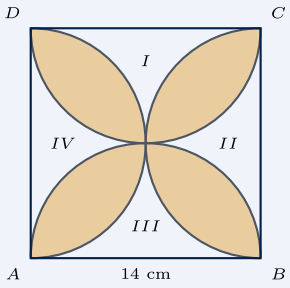
Q12. In a square $ABCD$ of side 14 cm, four semicircles are drawn with each side as diameter, intersecting to form a four-petaled central flower. **[5]**

(a) Label the four unshaded corner regions I, II, III, IV and derive the area of these unshaded regions.

(b) Calculate the total area of the four shaded petals. $\left(\text{Use } \pi = \frac{22}{7}\right)$

(c) Find the perimeter of the four shaded petals.

► STEP-BY-STEP SOLUTION: [5 Marks]



(a) Unshaded Corner Regions I, II, III, IV [2 Marks]

Area of square $ABCD = 14 \times 14 = 196$ cm^2. Each semicircle has diameter 14 cm $\implies r = 7$ cm.

$$\text{Area of 2 semicircles on } AB \text{ and } CD = 2 \times \left(\frac{1}{2}\pi \times 7^2\right) = 154 \text{ cm}^2$$

$$\text{Area}(I + III) = 196 - 154 = 42 \text{ cm}^2$$

Similarly, using semicircles on AD and BC: Area$(II + IV) = 196 - 154 = 42$ cm^2.

$$\text{Total unshaded area } (I + II + III + IV) = 42 + 42 = 84 \text{ cm}^2$$

(b) Area of Four Shaded Petals [2 Marks]

Area of petals = Area(square)$-$Total unshaded area $= 196 - 84 = 112$ cm^2

(c) Perimeter of Four Shaded Petals [1 Mark]

The perimeter consists of the arcs of the 4 semicircles:

$$\text{Perimeter} = 4 \times (\pi r) = 4 \times \left(\frac{22}{7} \times 7\right) = 4 \times 22 = 88 \text{ cm}$$

Unshaded Area $= \mathbf{84}$ cm$^\mathbf{2}$, Petal Area $= \mathbf{112}$ cm$^\mathbf{2}$, Perimeter $= \mathbf{88}$ cm

Q13. Botanical Garden Semicircular Flower Beds on Lawn

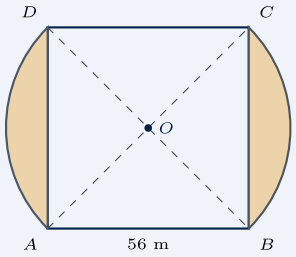
A municipal botanical park features a central square green lawn $ABCD$ of side length 56 m. On two opposite sides, circular sector flower beds are constructed, each centered at the diagonal intersection point O of the lawn.

(i) Find the distance from the lawn centre O to any corner vertex. [1]

(ii) Calculate the area of the central square green lawn. [1]

(iii) Calculate the combined total area of the lawn and both floral beds, **OR** calculate the area of the two floral beds alone. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [2]

▶ **STEP-BY-STEP SOLUTION:** [4 Marks]



(i) Distance from O to Any Vertex [1 Mark]

Side of square $a = 56$ m. Diagonal $d = a\sqrt{2} = 56\sqrt{2}$ m.

$$\text{Distance } OA = OB = OC = OD = r = \frac{d}{2} = \frac{56\sqrt{2}}{2} = 28\sqrt{2} \text{ m}$$

(ii) Area of Central Square Lawn [1 Mark]

$$\text{Area(square } ABCD) = a^2 = 56^2 = 3136 \text{ m}^2$$

(iii) Combined Total Area (or Area of Beds Alone) [2 Marks]

Diagonals of square intersect at right angles: $\angle AOD = \angle BOC = 90°$.

$$\text{Area(sector } OAD) = \frac{90°}{360°} \times \frac{22}{7} \times (28\sqrt{2})^2 = \frac{1}{4} \times \frac{22}{7} \times 1568 = 1232 \text{ m}^2$$

Similarly, Area(sector OBC) $= 1232$ m^2.

$$\text{Area}(\triangle OAB) = \text{Area}(\triangle OCD) = \frac{1}{4} \times 3136 = 784 \text{ m}^2$$

Total Combined Area:

$$\text{Total Area} = 1232 + 1232 + 784 + 784 = 4032 \text{ m}^2$$

OR (Area of two floral beds alone):

$$\text{Area(each bed)} = \text{Area(sector)} - \text{Area}(\triangle) = 1232 - 784 = 448 \text{ m}^2$$

$$\text{Total Area of both beds} = 2 \times 448 = 896 \text{ m}^2$$

(i) $28\sqrt{2}$ m, (ii) 3136 m^2, (iii) Combined: 4032 m^2 (OR Beds Alone: 896 m^2)

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